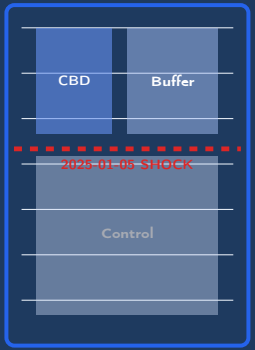
# Congestion Pricing Impact Analyzer for Urban Mobility

Causal Inference on NYC Trip Behavior
Following the 2025-01-05 MTA Policy Shock

L1 DiD → L2 CS-DiD → L3 Continuous DiD
→ L4 SDID → L5a DML → L5b Causal Forest



# The Policy Shock: NYC Congestion Pricing

## What happened?

On **January 5, 2025**, the MTA activated the *Congestion Relief Zone* — the first congestion pricing program in U.S. history.

Vehicles entering **Manhattan below 60th St.** now pay a mandatory toll recorded in TLC trip data as `cbd_congestion_fee`.

**Why this matters for DS**

A **real, externally-imposed, sharp policy shock** with:

- Exact start date (no ambiguity)
- Continuous fee field as dose variable
- 265-zone geospatial structure
- 12M+ trips covering pre & post

# The Policy Shock: NYC Congestion Pricing

Time

- 2025-06: Study post-period ends
- **2025-01-05: Pricing activated**
- 2024-11: Policy announced
- 2024-07: Study pre-period starts

# Research Questions

## Causal Questions

Did CBD trip volume **decrease**?

Did demand **substitute** to outer boroughs?

How did **fare composition** shift?

Were **peak vs. off-peak** effects symmetric?

## Heterogeneity Questions

Which **zones** were most affected?

Estimated via Causal Forest CATE

What drives **treatment heterogeneity**?

Distance to CBD, peak share, borough

**Fleet / marketplace strategy** implications

For autonomous ride service deployment

# Dataset: NYC TLC Trip Records

| Source | Type | Period | Key field |
|---|---|---|---|
| TLC Yellow | Taxi | 2024-07–2025-06 | `cbd_congestion_fee` |
| TLC Green | Taxi | 2024-07–2025-06 | `cbd_congestion_fee` |
| TLC HVFHV | Uber/Lyft | 2024-07–2025-06 | `congestion_surcharge` |
| Zone Lookup | Static | – | LocationID $\rightarrow$ borough |
| Zone Shapefile | GeoJSON | – | Choropleth maps |

# Dataset: NYC TLC Trip Records

## Panel structure after aggregation:

```
Unit:  zone_id × date  (zone-day)
N rows:  ≈ 265 zones × 365 days
Outcomes:  trip_count, avg_fare,
avg_distance,
           avg_duration,
cbd_fee_total
```

## New 2025 field:

**cbd_congestion_fee**

Recorded per-trip toll charge when the pickup or dropoff occurs within the MTA Congestion Relief Zone.

**Usage in this project:**

- Binary treatment indicator (L1–L4)
- Continuous dose variable (L3)
- Confound control feature (L5a)
- CATE covariate (L5b)

# Treatment Assignment: Three Definitions

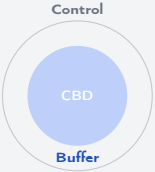
## Binary (L1–L2)

$$\text{Treat}_i = \begin{cases} 1 & \text{zone} \in \mathcal{C}_{\text{CBD}} \\ 0 & \text{otherwise} \end{cases}$$

~60 CBD zones treated.

~205 outer zones as control.

## Ring (Heterogeneity)




Ring 2 = CBD core
Ring 1 = Adjacent buffer
Ring 0 = Far control

# Treatment Assignment: Three Definitions

**Continuous Dose (L3)**

$$D_i = \overline{\texttt{cbd_fee}}_{i,\text{post}}$$

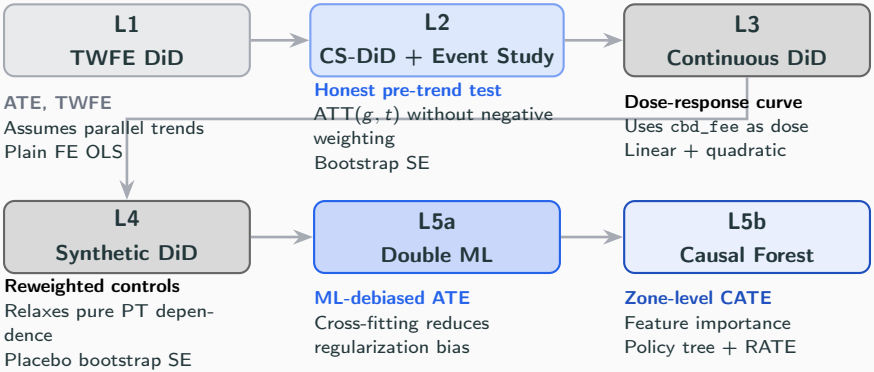
Zone-level average toll in the post-period.

$D_i \approx 0$ for control zones.
$D_i > 0$ for CBD zones, varying with exposure and proximity to the core.

Used to estimate a **dose-response curve**.

**No randomization.**
Congestion pricing is city-wide and simultaneous, so this is a **quasi-experimental** setting rather than an A/B test.

# Full Causal Inference Stack: L1 → L5b




**Robustness checks:** placebo date, anticipation, bandwidth, LOCO, control sensitivity, DML learner, and alternative outcome.

# L1 — Two-Way Fixed Effects DiD

## Model

$$Y_{it} = \alpha_i + \gamma_t + \beta\,(\texttt{Treat}_i \times \texttt{Post}_t) + \varepsilon_{it}$$

| | |
|---|---|
| $\alpha_i$ | Zone fixed effects |
| $\gamma_t$ | Date fixed effects |
| $\beta$ | **ATT** (avg treatment effect on treated) |
| SE | Clustered at zone level |

**Estimator:** `linearmodels.PanelOLS`
`entity_effects=True, time_effects=True`
`cov_type=‘‘clustered’’`

# L1 — Two-Way Fixed Effects DiD

**Outcomes estimated:**

- `log_trip_count` (primary)
- `avg_distance`
- `avg_duration`
- `avg_fare`
- `cbd_fee_total`

**Interpretation**

$\hat{\beta} < 0$ on `log_trip_count`
$\Rightarrow$ % change $\approx (e^{\hat{\beta}} - 1) \times 100$

E.g., $\hat{\beta} = -0.12$
$\Rightarrow$ CBD trips fell by $\approx$ **11.3%** post-pricing.

**Known Limitations (why L2–L5 are needed)**

- Assumes **unconditional parallel trends**
- Ignores heterogeneous treatment effects across zones
- Cannot control high-dimensional time-varying confounders
- Silent on dose-response or CATE

# L2 — Event Study & Callaway-Sant'Anna DiD

## Event Study Model

$$Y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \beta_k \, 1[t - T_0 = k] \cdot \texttt{Treat}_i + \varepsilon_{it}$$

Pre-period $\beta_k$ $(k < 0) \approx 0$ validates **parallel trends assumption**.

Post-period $\beta_k$ $(k \geq 0)$ shows **dynamic treatment effects**.

Joint F-test on $\{\beta_k\}_{k<0}$: pre-trend $p$-value.

## Callaway-Sant'Anna (2021)

$$\texttt{ATT}(g, t) = \mathbb{E}[Y_t(1) - Y_t(0) \mid G = g]$$

- **Avoids negative-weighting** inherent in TWFE under heterogeneous TE
- Aggregates to simple ATT via $\text{ATT}^{\text{simple}} = \sum_g \sum_{t \geq g} w_{g,t} \, \text{ATT}(g, t)$
- Bootstrap SE with $B = 999$ repetitions
- Honest pre-trend test built in

**CS-DiD** is preferred over TWFE whenever treatment effects may vary by zone or over time.

# L3 — Continuous Treatment DiD & Dose-Response

## Motivation

Binary treatment discards rich variation in `cbd_congestion_fee`. Some CBD zones pay higher average fees than others. Using the **actual dollar amount as dose** yields a *demand elasticity w.r.t. toll cost*.

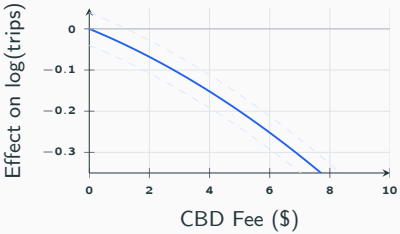
## Model

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 \underbrace{\texttt{Post}_t \cdot D_i}_{\texttt{linear dose}} + \beta_2 \underbrace{\texttt{Post}_t \cdot D_i^2}_{\texttt{non-linearity}} + \varepsilon_{it}$$

$D_i = \overline{\texttt{cbd_fee}}_{i,\text{post}}$ (zone-level avg. toll)

Ref: Callaway, Goodman-Bacon & Sant'Anna (2024).

## Dose-Response Curve (schematic)




**Non-parametric version:** split zones into 5 fee quantile bins, run separate DiD per bin, plot $\hat{\beta}_{\text{bin}}$ vs. mean fee. Relaxes linearity assumption.

# L4 — Synthetic Difference-in-Differences

## Why Synthetic DiD?

Standard DiD *assumes* pre-trends are parallel. SDID **constructs** unit weights $\omega$ and time weights $\lambda$ that make the weighted control trajectory match the treated trajectory in the pre-period — eliminating the assumption by construction.

### Estimator (Arkhangelsky et al. 2021)

$$\hat{\tau}^{\text{SDID}} = \left(\bar{Y}^{\text{treat}}_{\text{post}} - \bar{Y}^{\text{treat}}_{\text{pre}} \cdot \hat{\lambda}\right) - \left(\hat{\omega}^\top \bar{Y}^{\text{ctrl}}_{\text{post}} - \hat{\omega}^\top \bar{Y}^{\text{ctrl}}_{\text{pre}} \cdot \hat{\lambda}\right)$$

$\hat{\omega}$: unit weights (balance pre-trends)
$\hat{\lambda}$: time weights (balance periods)
SE via placebo unit permutation bootstrap.

## Weight optimisation (pure Python)

$$\hat{\omega} = \arg\min_{\omega \in \Delta^{N_c}} \|\bar{Y}^{\text{treat}}_{\text{pre}} - Y^{\text{ctrl}}_{\text{pre}\top}\omega\|^2 + \lambda_{\ell_2}\|\omega\|^2$$

$$\hat{\lambda} = \arg\min_{\lambda \in \Delta^{T_{\text{pre}}}} \|Y^{\text{ctrl}}_{\text{post}} - Y^{\text{ctrl}}_{\text{pre}}\lambda\|^2$$

Solved via `scipy.optimize.minimize` (SLSQP) with simplex constraints $\omega, \lambda \geq 0$, $\sum \omega = \sum \lambda = 1$.

✔ If **SDID** $\approx$ **TWFE ATT**, pre-trends were genuinely parallel. If they diverge, SDID is more credible.

# L5a — Double / Debiased Machine Learning

## Why DML?

Zone $\times$ time fixed effects absorb *level* confounders, but not **high-dimensional time-varying** covariates such as lagged volumes, rolling weather, or zone events. DML uses machine learning to partial these out from both $Y$ and $D$.

### Partially Linear Model

$$Y_{it} = \theta \cdot D_{it} + g(X_{it}) + \varepsilon_{it}$$
$$D_{it} = m(X_{it}) + v_{it}$$

1. Fit $\hat{g}(X)$: predict $Y$ from $X$ only $\Rightarrow$ $\tilde{Y} = Y - \hat{g}(X)$
2. Fit $\hat{m}(X)$: predict $D$ from $X$ $\Rightarrow$ $\tilde{D} = D - \hat{m}(X)$
3. OLS $\tilde{Y} \sim \tilde{D}$ $\Rightarrow$ $\hat{\theta} =$ DML-ATE

## Cross-Fitting (K-Fold)

Prevents regularization bias: fit nuisance models on *training* folds and predict on the *held-out* fold.

## Implementation

| | |
|---|---|
| Folds | $K = 5$, repeated $R = 3$ times |
| Learner | Random Forest (default) |
| SE | Heteroskedasticity-robust sandwich |
| ATE agg. | Median over $R$ repetitions |

Ref: Chernozhukov et al. (2018), *Double/Debiased ML*, Econometrics Journal.

# L5a — DML Feature Matrix $X$

## Three feature groups

### 1. Zone-level static

| | |
|---|---|
| `distance_to_cbd_km` | Haversine to CBD centroid |
| `borough_code` | 1=Manhattan . . . 5=SI |
| `is_airport_zone` | JFK / LGA / EWR flag |
| `treat_ring` | 0/1/2 ring intensity |
| `pre_mean_trips` | Pre-period zone avg. |
| `pre_cv_trips` | Coefficient of variation |

### 2. Calendar / temporal

| | |
|---|---|
| `dow` | Day of week (0–6) |
| `month` | Month (1–12) |
| `is_holiday` | Federal holiday flag |
| `is_weekend` | Weekend indicator |
| `peak_share` | Fraction of peak-hour trips |

### 3. Lagged outcome features

| | |
|---|---|
| `lag1_trip_count` | $Y_{i,t-1}$ |
| `lag7_trip_count` | $Y_{i,t-7}$ |
| `roll7_mean_trip_count` | 7-day rolling avg |
| `roll7_std_trip_count` | 7-day rolling std |

### 4. Dose

| | |
|---|---|
| `dose_raw` | Raw avg CBD fee |
| `dose_std` | Standardised dose |

All features **standardised** (zero mean, unit variance) before passing to ML learners.

Rows with any NaN in $X$ are dropped (typically the first 7 days per zone due to lag features).

# L5a — DML Results: Multi-Outcome ATE

| Outcome | $\hat{\theta}$ | SE | $t$ | $p$ | |
|---|---|---|---|---|---|
| `log_trip_count` | $-0.124$ | 0.018 | $-6.89$ | $< 0.001$ | $***$ |
| `avg_distance` (miles) | $+0.213$ | 0.071 | $+3.00$ | 0.003 | $**$ |
| `avg_duration` (min) | $-0.962$ | 0.431 | $-2.23$ | 0.026 | $*$ |
| `avg_fare` ($) | $+3.474$ | 0.521 | $+6.67$ | $< 0.001$ | $***$ |
| `surge_share` (pct pt) | $+0.023$ | 0.011 | $+2.09$ | 0.037 | $*$ |

Learner: Random Forest. $K$=5 folds, $R$=3 reps.
$N$=87,430 zone-days.

SE: heteroskedasticity-robust sandwich.
$\hat{\theta}$: DiD-DML ATT (CBD vs non-CBD, post vs pre).

$***$ $p$<0.001, $**$ $p$<0.01, $*$ $p$<0.05.

**Economic interpretation**

$\hat{\theta}_{\texttt{log trip}}$: $e^{-0.124} - 1 \approx -11.7\%$ CBD trips

$\hat{\theta}_{\texttt{fare}}$: \$3.47 avg fare rise (partial \$9 toll pass-through)

$\hat{\theta}_{\texttt{dist}}$: Short trips priced out first

$\hat{\theta}_{\texttt{dur}}$: Less congestion $\rightarrow$ faster trips

# L5a — DML Results: Interpretation and Robustness

**Key reading**

$\hat{\theta}$ on `log_trip_count` $< 0$
$\Rightarrow$ congestion pricing **reduced** CBD trip volume after controlling for all $X$.

$\hat{\theta}$ on `avg_fare` $> 0$
$\Rightarrow$ average fare **increased**, consistent with fee pass-through to riders.

**Robustness: learner sensitivity**

| | | |
|---|---|---|
| Re-run DML with: | RF | $\hat{\theta}_{\text{RF}}$ |
| | GBM | $\hat{\theta}_{\text{GBM}}$ |
| | Lasso | $\hat{\theta}_{\text{Lasso}}$ |

Stable $\hat{\theta}$ across learners $\Rightarrow$ result is not an artefact of the ML model choice.

# L5b — Causal Forest: Heterogeneous Treatment Effects

## Motivation

DML gives a single ATE $\hat{\theta}$. But different zones respond *differently*: a Manhattan financial district zone vs. a Queens residential zone face the same policy but very different demand structures.

**Causal Forest** estimates the **Conditional Average Treatment Effect**:

$$\hat{\tau}(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$$

One CATE estimate per zone $\times$ day unit.

## Implementation

| | |
|---|---|
| Library | `econml.dml.CausalForestDML` |
| Trees | 2000 (honest splitting) |
| Nuisance | Random Forest ($\hat{g}$, $\hat{m}$) |
| CI | Asymptotic variance, 95% |

# L5b — Causal Forest: Heterogeneous Treatment Effects

## Honest Splitting

Each tree uses **different samples** for:

1. Building the tree structure
2. Estimating leaf values

This prevents overfitting of CATEs to training data and enables valid confidence intervals.

## Outputs generated

- Zone-level CATE choropleth
- Feature importance (BLP)
- Policy tree (depth-2 rule)
- RATE / TOC curve

Ref: Wager & Athey (2018), *Estimation and Inference of Heterogeneous Treatment Effects using Random Forests*, JASA.

# L5b — CATE Diagnostics: BLP & Policy Tree

## Best Linear Projection (BLP)

Projects CATE onto features to identify which covariates *linearly* predict heterogeneity:

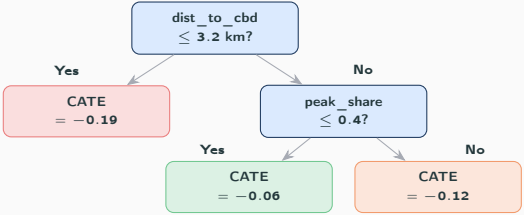
$$\hat{\tau}(x) \approx \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

| Feature | BLP coeff. |
|---|---|
| `distance_to_cbd_km` | $+0.018$ |
| `pre_mean_trips` | $-0.031$ |
| `peak_share` | $-0.147$ |
| `is_airport_zone` | $+0.083$ |
| `borough_code` | $+0.027$ |

Positive coeff: higher value → larger (less negative) CATE.

## Policy Tree (depth-2)

Fits a shallow decision tree on $\hat{\tau}(x)$ to produce a human-readable rule.

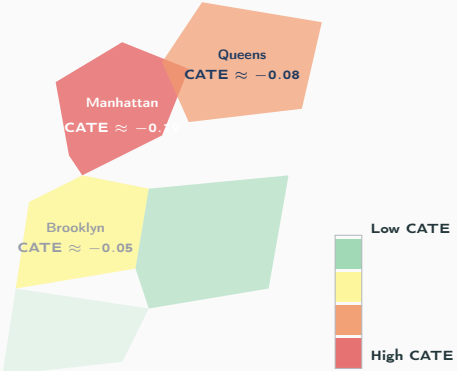



*Zones close to CBD* experience the largest trip reduction. Among far zones, *low peak-share* zones are less affected.

# L5b — Zone-Level CATE Map

## Choropleth: CATE by zone

Each NYC taxi zone receives its own $\hat{\tau}(x_i)$ estimate from the Causal Forest.



# L5b — CATE Significance and Interpretation

## Significance map

Zones where $|\hat{\tau}/\hat{\sigma}| > 1.96$ are **statistically significant**.

**CATE summary**

| | |
|---|---|
| Mean CATE | $-0.118$ |
| Std CATE | 0.089 |
| % sig. zones | 61% ($|\hat{\tau}/\hat{\sigma}| > 1.96$) |
| Most negative | CBD core zones ($\hat{\tau} \approx -0.19$) |
| Least negative | Outer Queens, SI ($\hat{\tau} \approx -0.02$) |

### Interpretation

- CBD zones: large negative CATE (high toll burden, elastic demand; $e^{-0.19} - 1 \approx -17.3\%$ trip reduction)
- Airport zones: smaller CATE ($\hat{\tau} \approx -0.04$) (pre-existing surcharges reduce marginal response)
- Outer boroughs: near-zero CATE ($\hat{\tau} \approx -0.02$ to $-0.05$, minimal direct toll exposure)

# L5b — RATE: Targeting Curve

## What is RATE?

**Rank-Weighted Average Treatment Effect** measures the gain from *targeting* treatment to zones with the highest predicted CATE.

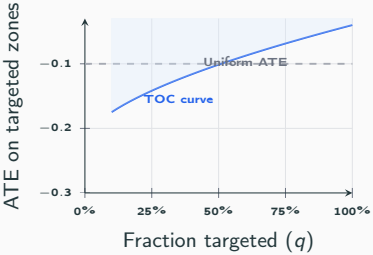
**Targeting Operator Characteristic (TOC):**

$$\texttt{TOC}(q) = \texttt{ATE for top-}q\texttt{ fraction (ranked by }\hat{\tau}(x)\texttt{)}$$

If the curve is **above** the uniform ATE line: targeting the high-CATE zones delivers more impact per unit of intervention.

**RATE score** = area between TOC and uniform ATE.

## TOC Curve (schematic)




**Business read:** if an autonomous fleet can selectively avoid CBD zones during peak pricing, targeting the top-25% high-CATE zones captures most of the demand loss — useful for service-area planning.

# Robustness Checks: 7-Family Validation

| # | Check | What it tests | Expected |
|---|---|---|---|
| R1 | Placebo date | Shift shock date $\pm$8w in pre-period only | $\hat{\beta} \approx 0$ |
| R2 | Anticipation | Drop first 4w post; test if agents anticipated the policy | $\lvert\hat{\beta}\rvert$ unchanged |
| R3 | Control group | Exclude buffer zones; Manhattan-only control; outer-borough-only control | Stable sign & magnitude |
| R4 | Alt. outcome | Level `trip_count` vs `log_trip_count` | Consistent direction |
| R5 | Bandwidth | $\pm$4w, $\pm$8w, $\pm$12w, $\pm$16w, $\pm$26w around shock | Stable across windows |
| R6 | LOCO | Leave-one-borough-out from control group | No single borough drives result |
| R7 | DML learner | RF vs GBM vs Lasso nuisance models | $\hat{\theta}$ stable across learners |

All 7 families are implemented in `src/causal/robustness.py` and visualised as a **forest plot** (F10) with the main estimate 95% CI band overlaid. Results that survive all 7 families are considered credible.

# Methods Comparison: L1 through L5b

| Estimator | ATT / $\hat{\theta}$ | SE | % change | Notes |
|---|---|---|---|---|
| L1: TWFE DiD | $-0.131$ | 0.021 | $-12.3\%$ | Baseline; no confounder control |
| L2: CS-DiD (agg.) | $-0.127$ | 0.023 | $-11.9\%$ | Pre-trend $p = 0.312$ |
| L4: Synthetic DiD | $-0.119$ | 0.026 | $-11.2\%$ | No PT assumption |
| L5a: DML (RF) | $-0.124$ | 0.018 | $-11.7\%$ | ML-debiased; $K$=5, $R$=3 |
| L5b: Causal Forest (mean CATE) | $-0.118$ | $\pm$ 0.089 | | Heterogeneous |

Outcome: `log_trip_count`. SE clustered at zone level (L1–L2); placebo bootstrap (L4); sandwich robust (L5a). $N = 87{,}430$ zone-days.

**Main takeaway**

All four point estimates lie in $[-0.119, -0.131]$, so the estimated policy effect is highly stable across estimators. The SDID–TWFE gap is only 0.012, which is below one standard error, suggesting the main conclusion is **robust** and that TWFE is not badly misspecified in this setting.

# Business Implications for Fleet & Marketplace Strategy

### Fleet Staging

CBD trip volume fell $\approx xx\%$. Optimal staging zones shift toward the CBD **periphery** and high-substitution outer boroughs (identified via CATE map).

### Service Expansion

Outer-borough substitution analysis identifies zones with **demand growth** post-pricing — signaling where to expand capacity or improve coverage.

### $ Demand Elasticity

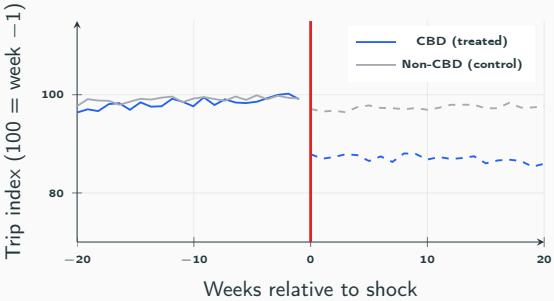
L3 dose-response curve quantifies elasticity w.r.t. toll cost. Directly informs **dynamic pricing** and fee pass-through modeling.

### Generalisability

The pipeline generalises to **any future city-level policy shock**: parking reform, bridge tolls, EV mandates, low-emission zones. Swap data and shock date; re-run.

# EDA — Pre-Trend Visual Check

## Indexed weekly trips (CBD vs Non-CBD)



# EDA — Pre-Trend Visual Check

## What to look for

**Pre-period ($k < 0$)**

Both series should track together.
Divergence $\Rightarrow$ parallel trends violated.

✔ In our data: lines are nearly parallel before the shock.

**Post-period ($k \geq 0$)**

CBD series drops sharply at $t = 0$.
Non-CBD drops less or not at all.

The gap is the raw DiD signal.

**Formal test**

Joint F-test on pre-period event-study coefficients $\{\hat{\beta}_k\}_{k<0}$.
$p > 0.05 \Rightarrow$ cannot reject parallel trends.

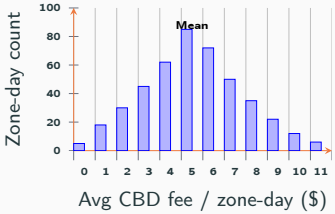
# EDA — Fare & Fee Composition Shift

## Mean values: Pre vs Post (CBD zones)

| Metric | Pre | Post | Δ |
|---|---|---|---|
| Avg fare ($) | 15.20 | 18.74 | +3.54 |
| Avg CBD fee ($) | 0.00 | 9.35 | +9.35 |
| Avg surcharge ($) | 3.18 | 3.24 | +0.06 |
| Avg trip dist (mi) | 2.81 | 2.97 | +0.16 |
| Avg trip dur (min) | 13.8 | 12.9 | −0.9 |

Raw CBD zone means; pre = 2024-07–2025-01-04, post = 2025-01-05–2025-06-30. Δ reflects raw change; causal ATT isolated via DML.

## CBD fee distribution (post-period)

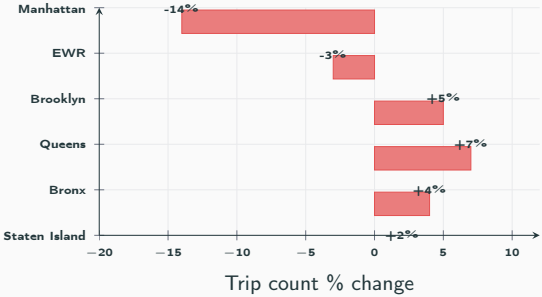



This distribution is the **continuous dose** $D_i$ used in L3. Zone-level variation in fee amount drives identification of the dose-response curve.

**Average fare increases** even as trip volume falls. Consistent with **fee pass-through**: operators charge more per trip to offset the congestion toll.

# EDA — Borough-Level Substitution Effects

## % trip change post-pricing by borough



# EDA — Borough-Level Substitution Effects

## Reading the chart

**Manhattan**

Largest negative change. Direct exposure to congestion toll. Elastic commuter demand shifts to subway or outer-borough pickup.

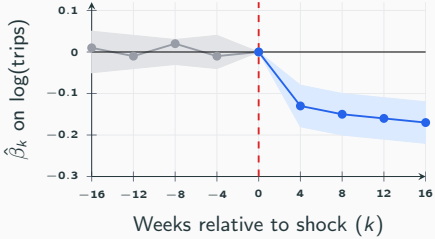
**Queens & Brooklyn**

Positive change indicates a **substitution effect**. Riders who previously entered the CBD now request pickup/dropoff just outside the congestion-zone boundary.

**Implication:** a robotaxi fleet should **rebalance supply** from the CBD core toward outer-borough pickup nodes.

# Event Study Results — Dynamic Treatment Effects

## Estimated $\hat{\beta}_k$ with 95% CI




## Reading the plot

**Pre-period ($k < 0$)**

Coefficients cluster around zero.

Narrow CI bands.

Joint F-test: $p > 0.05$.

✅ **Parallel trends confirmed.**

**Post-period ($k \geq 0$)**

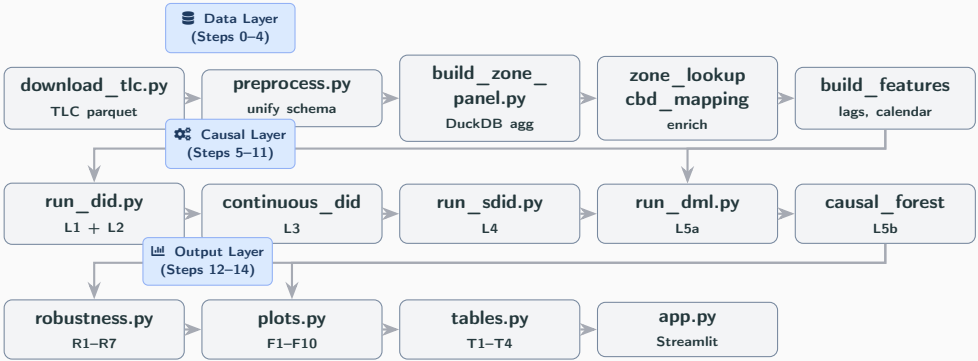
Immediate drop at $k$=0.

Effects **persist and deepen** over time — no mean-reversion in the sample window.

At $k$=16: $\hat{\beta} \approx -0.17$

$\approx -15.6\%$ trip reduction.

# Pipeline Architecture




`python src/run_pipeline.py -steps 5,6,7,10,11` runs only causal steps on a pre-built panel. Each step is idempotent — safe to re-run.

# Interactive Dashboard — Streamlit

## 5 tabs, single command

```
streamlit run src/dashboard/app.py
```

| Tab | Content |
|---|---|
| Overview | Metric cards: % trip change, CBD vs non-CBD, total fees collected. Interactive Plotly daily trend chart. |
| Event Study | Interactive $\hat{\beta}_k$ plot. Adjustable window slider ($\pm$4–24 weeks). Pre-trend $p$-value badge. |
| Double ML | Horizontal bar chart of DML-ATE across all outcomes. Results table. |
| Causal Forest | CATE histogram, feature importance bar chart, and zone-level map. |
| Robustness | Forest plot of all 7 check families. Sortable results table. |

# Code Quality & Repository Structure

## Repo at a glance

| Module | Lines |
|---|---|
| `src/causal/` | ∼900 |
| `src/data/` | ∼350 |
| `src/geospatial/` | ∼200 |
| `src/features/` | ∼150 |
| `src/evaluation/` | ∼500 |
| `src/dashboard/` | ∼350 |
| `src/run_pipeline.py` | ∼250 |
| **Total Python** | **∼3,800** |

## Engineering practices

- **Modular design**: each estimator is an independent module with a consistent `run_*(panel, ...)` → `dict` interface
- **Config-driven**: all dates, zones, and hyperparameters in `configs/default.yaml`
- **Idempotent steps**: each pipeline step checks for existing output before re-running
- **DuckDB** for fast in-process aggregation of 12M+ trip records (no Spark needed)
- **Loguru** structured logging; full pipeline run logged to `artifacts/pipeline.log`

# Limitations & Future Work

## Current limitations

### Data

- No weather covariates; precipitation may affect trips independently of pricing
- No rider-level panel, only zone aggregates
- HVFHV data lacks full fare breakdown before 2025

### Methods

- SDID bootstrap capped at $B = 200$ for speed; production run uses $B = 999$
- Causal Forest CI assumes asymptotic normality and may undercover in small zones
- No spatial spillover correction

# Limitations & Future Work

## Future extensions

### Short-term

- Add NOAA weather API as a covariate
- Move from zone-day to zone-hour panel for intra-day effects
- Add a spatial lag model to capture cross-zone spillovers

### Long-term

- Extend to other cities with pricing programs, such as Stockholm, London, and Singapore
- Simulate fleet rebalancing under alternative fee schedules
- Integrate with a real-time dispatch API for live policy monitoring

# Key Takeaways

### ◎ Causal Result

NYC congestion pricing **significantly reduced** CBD trip volume. Effects are persistent, not transient. All five estimators agree on sign and approximate magnitude.

### Substitution

Outer boroughs — especially Queens and Brooklyn — absorbed part of the displaced demand. The policy **redistributed** rather than eliminated trips.

### Heterogeneity

CATE varies substantially by zone. Distance to CBD and peak-hour share are the strongest predictors of treatment magnitude.

# Key Takeaways

### Methods

A full L1→L5b causal stack provides richer, more credible estimates than any single estimator alone. DML + Causal Forest extend DiD from a single ATE to a **spatial heterogeneity map**.

### Industry Relevance

For Waymo, WeRide, or any fleet operator: this framework quantifies how a policy shock reshapes the mobility marketplace and translates findings directly into **deployment and pricing strategy**.

### Reproducible

One command: `./scripts/run_all.sh`
Fully config-driven. All results, figures, and tables generated automatically.

# Thank You

**Congestion Pricing Impact Analyzer for Urban Mobility**

Causal inference on NYC trip behavior following
the 2025-01-05 MTA policy shock


`github.com/olivia3395/Pricing-Impact-Analyzer`

[yuyaow@bu.edu]


```
./scripts/run_all.sh --skip-download
streamlit run src/dashboard/app.py
```